

Advanced Database

Chapter 06: Data Warehouse

PLAN

❖ **Data Warehouse**

- I. Introduction
- II. Terminologies
- III. OLAP
 - 1. OLAP Schemas
 - 2. OLAP Operations
- IV. Partitioning strategy
- V. Process Managers
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Introduction

- ❖ The term "Data Warehouse" was first coined by Bill Inmon in 1990. According to Inmon, a data warehouse is a subject oriented, integrated, time-variant, and non-volatile collection of data. This data helps analysts to take informed decisions in an organization.

Introduction

❖ Data Warehouse Features

■ Subject Oriented,

- ✓ These subjects can be product, customers, suppliers, sales, revenue, etc.
- ✓ A data warehouse does not focus on the ongoing operations, rather it focuses on modelling and analysis of data for decision making.

■ Integrated

- ✓ A data warehouse is constructed by integrating data from varied sources.

■ Time Variant

- The data collected in a data warehouse is identified with a particular time period. It contains "old" data, dating back several years, used for comparisons, forecasts, etc.

■ Non-volatile

- ✓ The previous data is not erased when new data is added to it.
- ✓ Once added in warehouse, data are not modified .

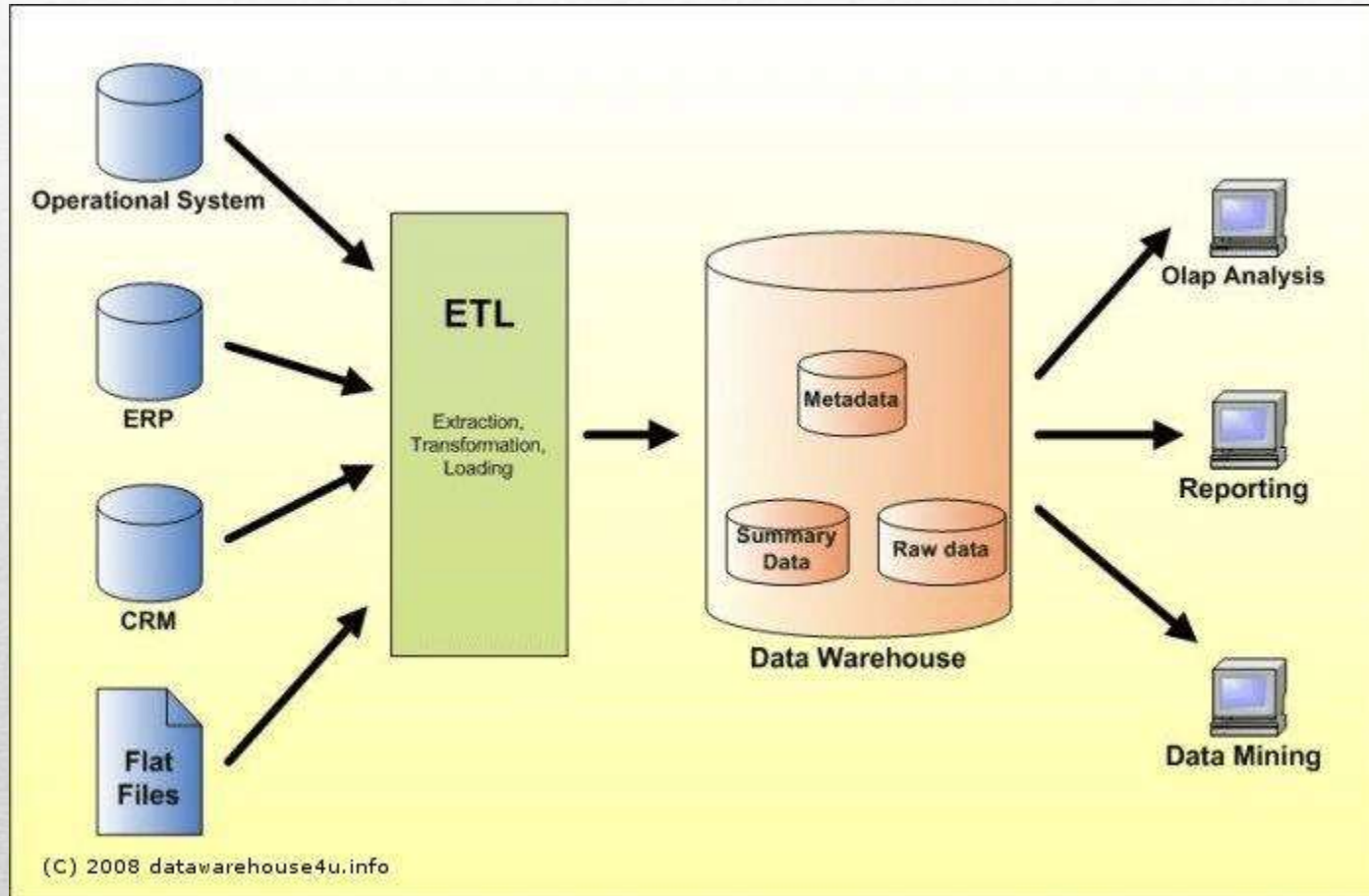
Introduction

❖ Necessity for a data warehouse

- Gather company data in one place without overloading the databases (operational systems)
- Provide access to various sources of data and ensure data quality
- Extract, filter, and integrate relevant information, for subsequent requests
- Gain knowledge and learn about the company, the market and the environment

Introduction

Architecture of a Data Warehouse



Introduction

❖ Data Warehouse Applications

- A data warehouse helps business executives to organize, analyze, and use their data for decision making.
- Data warehouses are widely used in : financial services, banking services (customer profile, ...), marketing, retail sectors, ...

❖ Benefits:

- From the data warehouse, various analyzes can be made, in particular by “On-Line Analytical Processing” (OLAP) or Data-Mining and Visualization.
- The information and knowledge obtained by exploitation of data warehouse have a direct impact on profits of the company (increased sales through marketing more targeted, improved stock rotation, etc.)

System process

- ❖ It describe the functions of the Data Warehouse
 - **Data Extraction** – Involves gathering data from multiple heterogeneous sources.
 - ✓ Data needs to be in a consistent state when it is extracted,
 - **Data Cleaning** – Involves finding and correcting the errors in data.
 - ✓ making the data consistent with other data
 - **Data Transformation** – Involves converting the data from legacy format to warehouse format.
 - **Data Loading** – Involves sorting, summarizing, consolidating, checking integrity, and building indices and partitions.
 - **Refreshing** – Involves updating from data sources to warehouse.

Database vs Data Warehouse

Database	Data Warehouse
Database contains detailed data.	data warehouse contains summarized data
Database uses Online Transactional Processing (OLTP)	Data warehouse uses Online Analytical Processing (OLAP).
Database helps to perform business operations	Data warehouse helps to analyze the business data.
Database is application oriented.	Data warehouse is subject oriented.
Tables and joins of a database are complex because they are normalized. This minimizes data redundancy.	Tables and joins are simple in data warehouse because they are de-normalized.
Limited to a single application	Draws data from a range of applications
Data within operational system are updated regularly	Data is non-volatile , when new data is added, old data is not erased

Terminologies

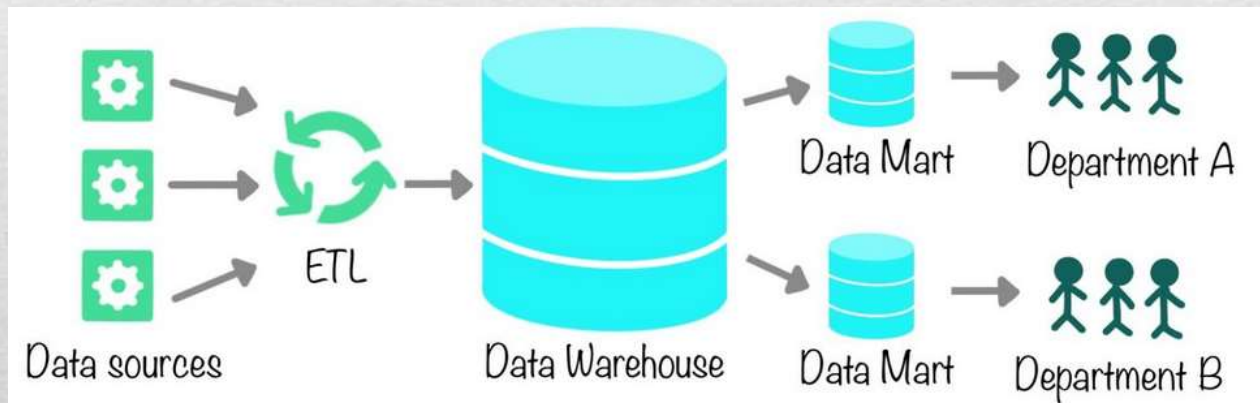
❖ Metadata

- Metadata is simply defined as data about data, the data that are used to represent other data (provenance, aggregation method used, ...)
 - ✓ Metadata is the road-map to data warehouse
 - ✓ Defines the warehouse objects
 - ✓ Acts as a directory, that helps the decision support system to locate the contents of a data warehouse
 - ✓ Metadata Repository has the following metadata:
 - **Definition of data warehouse** (structure, views, hierarchies, derived data, ...)
 - **Business metadata,**
 - **Operational Metadata,**
 - **Data for mapping from operational environment to data warehouse,**
 - **Algorithms for summarization**

Terminologies

❖ Data Mart

- Data marts contain a subset of organization-wide data that is valuable to specific groups of people in an organization.
- It stores the information of a particular function of an organization which is handled by single authority.



Terminologies

❖ Data mining

- The practice of analyzing the big data present in data warehouse is data mining.
- It is used to find the hidden patterns that are present in the database or in data warehouse with the help of algorithm of data mining.

OLAP

- ❖ OLAP is an acronym for **OnLine Analytical Processing**. OLAP performs multidimensional analysis of business data and provides the capability for complex calculations, trend analysis, and sophisticated data modeling.
- **Foundation for business applications:**
Business Performance Management, Planning, Budgeting, Forecasting, Financial Reporting, Analysis, Simulation Models, Knowledge Discovery, and Data Warehouse Reporting.



OLAP

❖ Nature of OLAP analysis

- Aggregation - (total sales, percent-to-total, ...)
- Comparison - budget vs expenses
- Ranking - Top 10,
- Access to detailed and aggregated data
- Specification of complex criteria

OLAP

❖ Difference between OLAP and OLTP in DBMS

- Online Analytical Processing (OLAP) - consists of a type of software tools that are used for data analysis for business decisions. OLAP provides an environment to get insights from the database retrieved from multiple database systems at one time.
 - ✓ **Example:** Any type of Data warehouse system is an OLAP system.
- Online transaction processing (OLTP) – provides transaction-oriented applications. OLTP administers day to day transaction of an organization.
 - ✓ **Example:** ATM center is an OLTP application.

OLAP

❖ Multidimensional modeling

■ Dimension:

- ✓ Presents the point of view according to which we want to see the data described by a set of attributes, Axis of the analysis
 - Example: Orders, purchases, complaints, products, clients,...

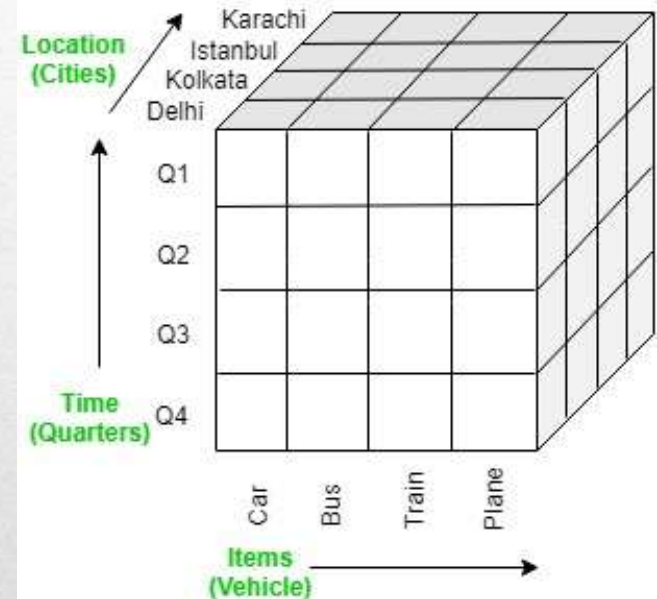
■ Measures / facts

- ✓ Numerical function which can be evaluated at any point of the data cube by aggregating the data corresponding to this point, measurement (analysis criterion)
 - Example sales, gain, ...

OLAP

❖ Hyper OLAP cube

- **Goals**
 - ✓ Get information already aggregated according to user needs
 - ✓ Representation of information in an N-dimensional hyper cube
- **OLAP operations**
 - ✓ Functionalities that are used to facilitate multidimensional analysis: operations that can be performed on the hyper cube



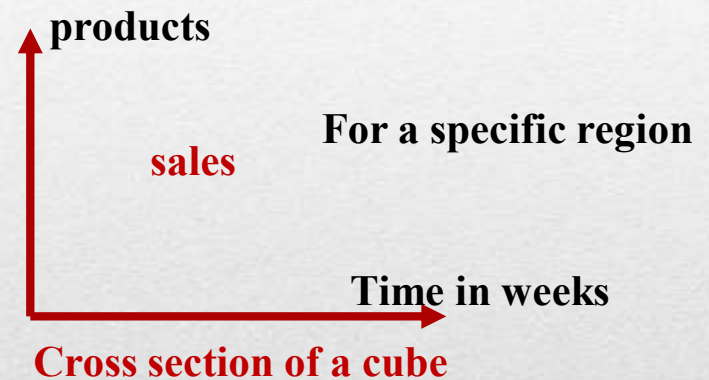
OLAP

❖ Granularity of dimensions

- **Geography:** Regions, countries, cities,
- **Products:** Types, categories, ranges, brands,
- **Time:** Years, quarters, Months, days,

OLAP

❖ multidimensional navigation



❖ OLAP vs. Data Mining

- OLAP and data mining are used to solve different kinds of analytical problems.
- OLAP summarizes data and makes forecasts.
 - ✓ Multidimensional views of data
- Data mining discovers hidden patterns in data and operates at a detailed level instead of a summary level.
 - ✓ Automatic Pattern Prediction based on trend and behavior analysis.
 - ✓ They involve ratios and algorithms like decision trees, nearest neighbor classification and neural networks,

Types of OLAP

- ROLAP: Relational OLAP
 - ✓ To handle multidimensionality of data, the data cube is stored as relational database.
 - ✓ ROLAP is used for large data volumes.
 - ✓ Performance of ROLAP can be slow.
- MOLAP: Multidimensional OLAP – Cube based
 - ✓ MOLAP stores data on disks in specialized multidimensional array structure.
 - ✓ Used for complex calculations.
 - ✓ Optimal for operation such as slice and dice.
 - ✓ Processing time of MOLAP is fast.
- Hybrid OLAP
 - ✓ Hybrid OLAP is a combination of both ROLAP and MOLAP. It offers higher scalability of ROLAP and faster computation of MOLAP.

OLAP Schemas

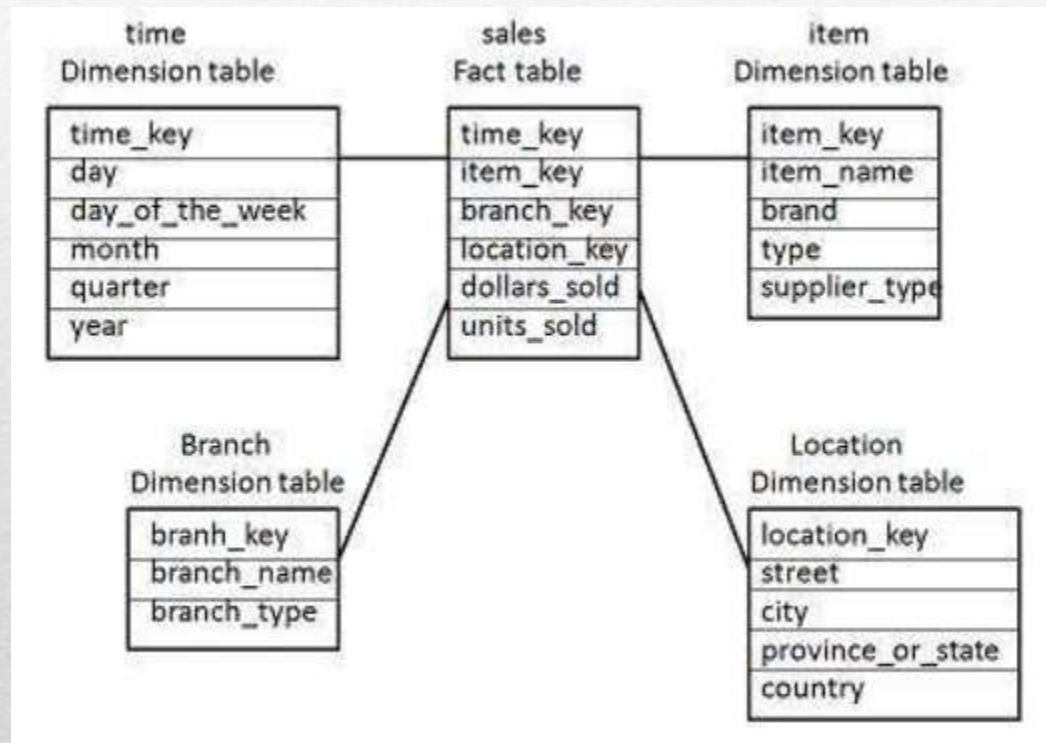
- Schema is a logical description of the entire database.
- Much like a database, a data warehouse also requires to maintain a schema.
- A database uses relational model, while a data warehouse uses Star, Snowflake, and Fact Constellation schema.

OLAP Schemas

❖ Star Schema

✓ Each dimension in a star schema is represented with only one-dimension table.

■ Example



- ✓ There is a fact table at the center. It contains the keys to each of four dimensions.
- ✓ The fact table also contains the attributes, namely dollars sold and units sold.

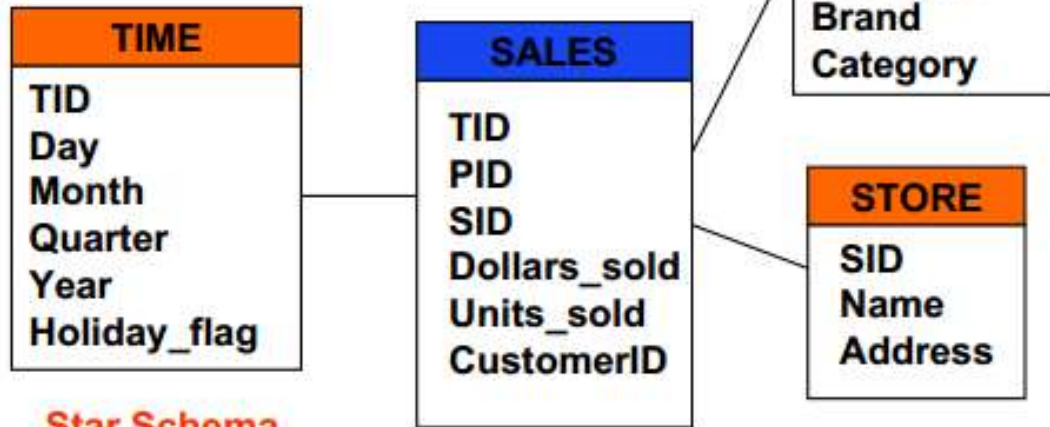
OLAP Schemas

❖ Star Schema – Example

```
SELECT  P.brand, sum(dollars_sold), sum(units_sold)
FROM    SALES S, PRODUCT P, TEMPS T
WHERE   S.PID = P.PID (Jointure)
AND     S.TID = T.TID (Jointure)
AND     T.Quarter = "1 Q 97" (Sélection)
GROUP BY P.brand
ORDER BY P.brand
```

Star Join Request

- Multiple joins
- Followed by selections

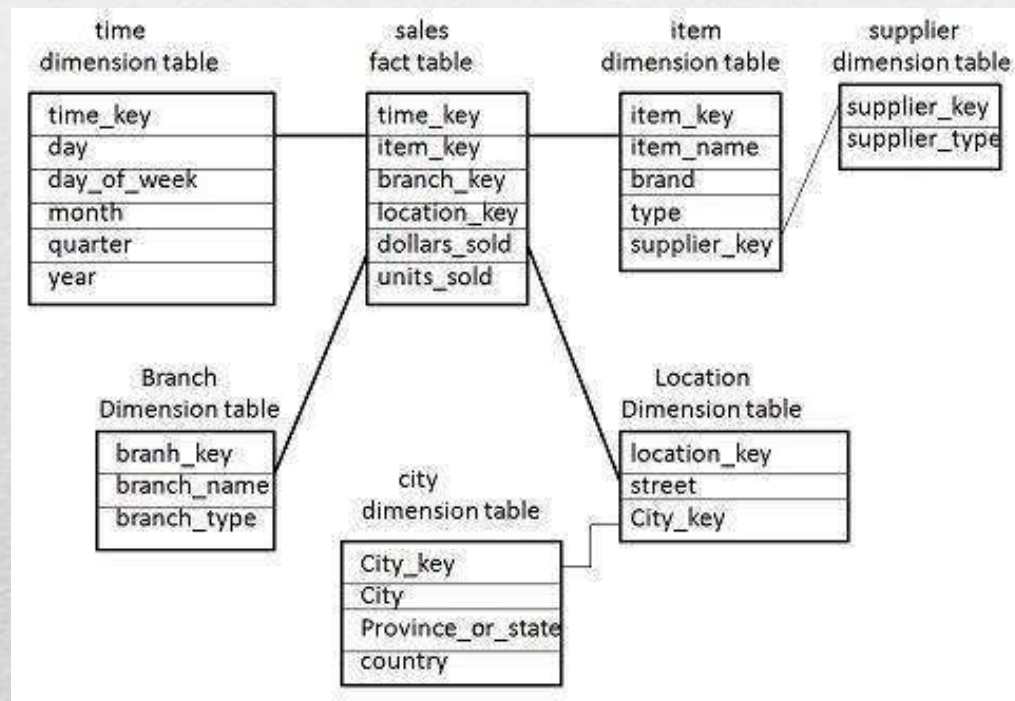


Star Schema

OLAP Schemas

❖ Snowflake Schema

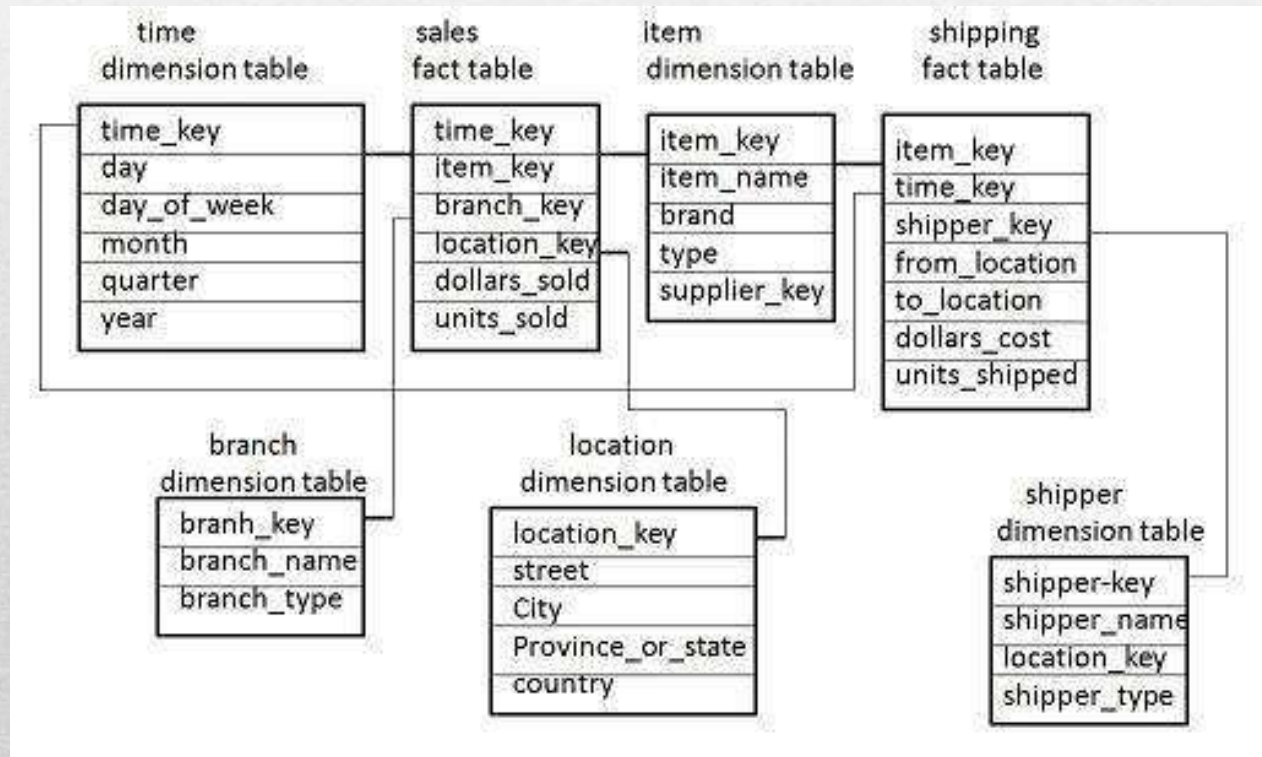
- ✓ Some dimension tables in the Snowflake schema are normalized.
- ✓ The normalization splits up the data into additional tables.
- ✓ **For example**, the item dimension table in star schema is normalized and split into two dimension tables, namely item and supplier table.



OLAP Schemas

❖ Fact Constellation Schema (galaxy schema)

- A fact constellation has multiple fact tables.
- The following diagram shows two fact tables, namely sales and shipping.



OLAP Schemas

❖ Schema definition

- ✓ Multidimensional schema is defined using Data Mining Query Language (**DMQL**).
- ✓ The two primitives, cube definition and dimension definition, can be used for defining the data warehouses and data marts.

■ Syntax for Cube Definition

```
define cube <cube_name> [<dimension-list >]: < measure_list >
```

■ Syntax for Dimension Definition

```
define dimension <dimension_name> as (<attribute_or_dimension_list> )
```

OLAP Schemas

- Star Schema Definition

- ✓ The star schema that we have discussed can be defined using Data Mining Query Language (DMQL) as follows :

```
define cube sales star [time, item, branch, location]:
```

```
dollars sold = sum(sales in dollars), units sold = count(*)
```

```
define dimension time as (time key, day, day of week, month, quarter, year)
```

```
define dimension item as (item key, item name, brand, type, supplier type)
```

```
define dimension branch as (branch key, branch name, branch type)
```

```
define dimension location as (location key, street, city, province or  
state, country)
```

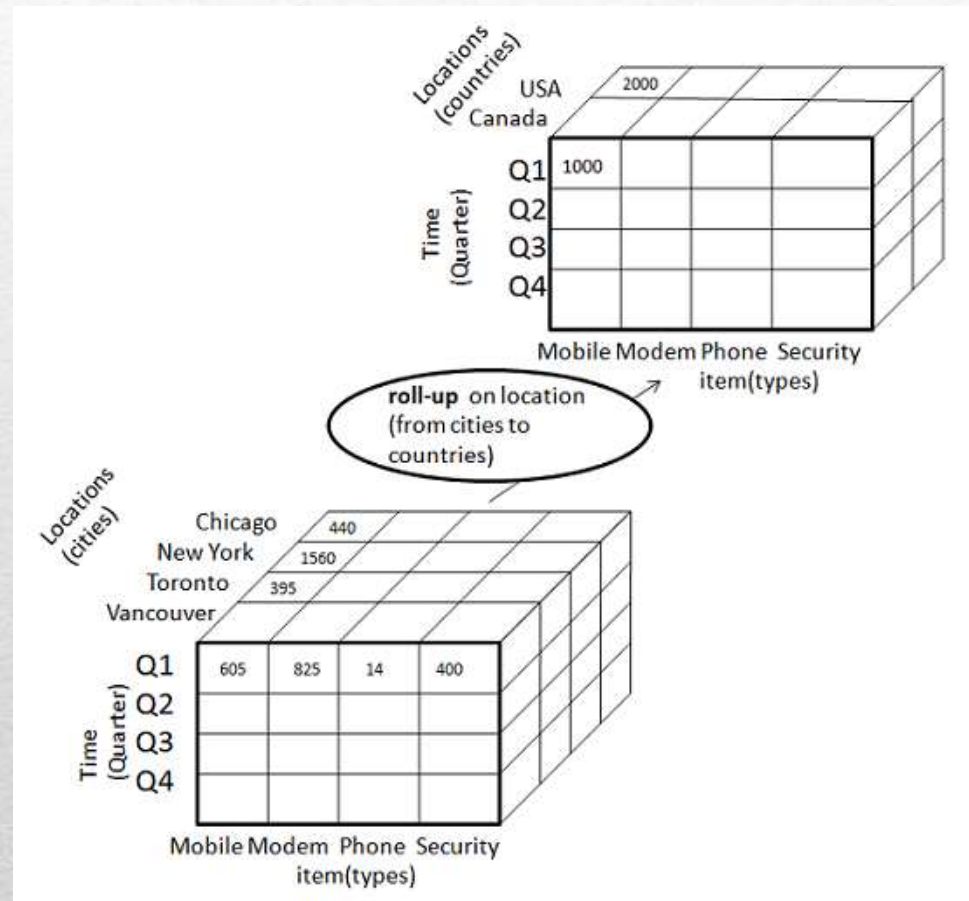
OLAP Schemas

❖ Snowflake Schema Definition

```
define cube sales snowflake [time, item, branch, location]:  
  
dollars sold = sum(sales in dollars), units sold = count(*)  
  
define dimension time as (time key, day, day of week, month, quarter,  
year)  
  
define dimension item as (item key, item name, brand, type, supplier  
(supplier key, supplier type))  
  
define dimension branch as (branch key, branch name, branch type)  
  
define dimension location as (location key, street, city (city key,  
city, province or state, country))
```

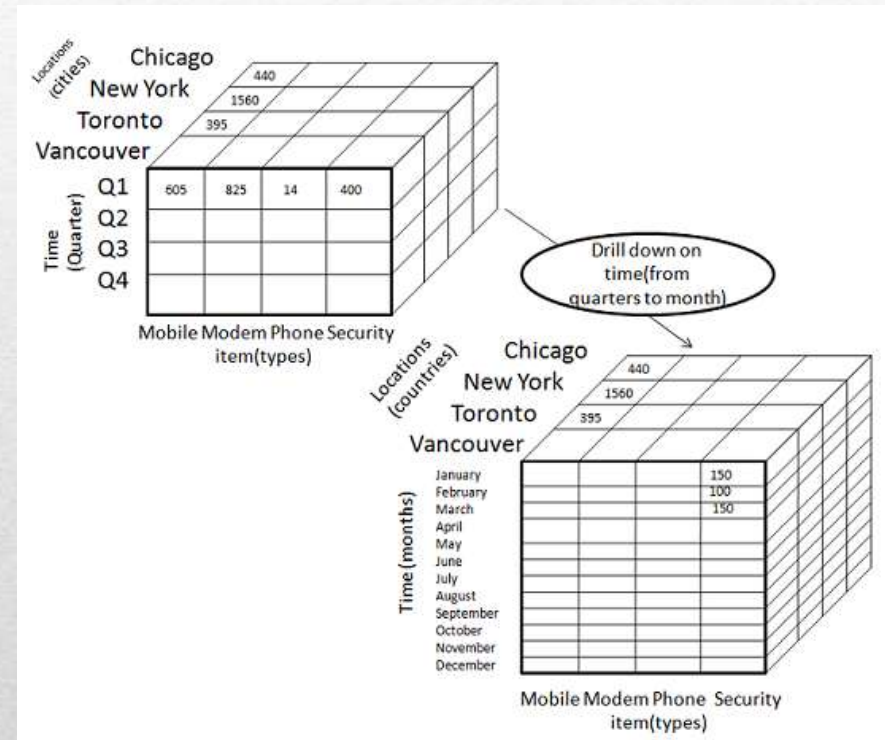
OLAP Operations

- **Roll-up**
 - ✓ Roll-up performs aggregation on a data cube in any of the following ways:
 - By climbing up a concept hierarchy for a dimension
 - By dimension reduction



OLAP Operations

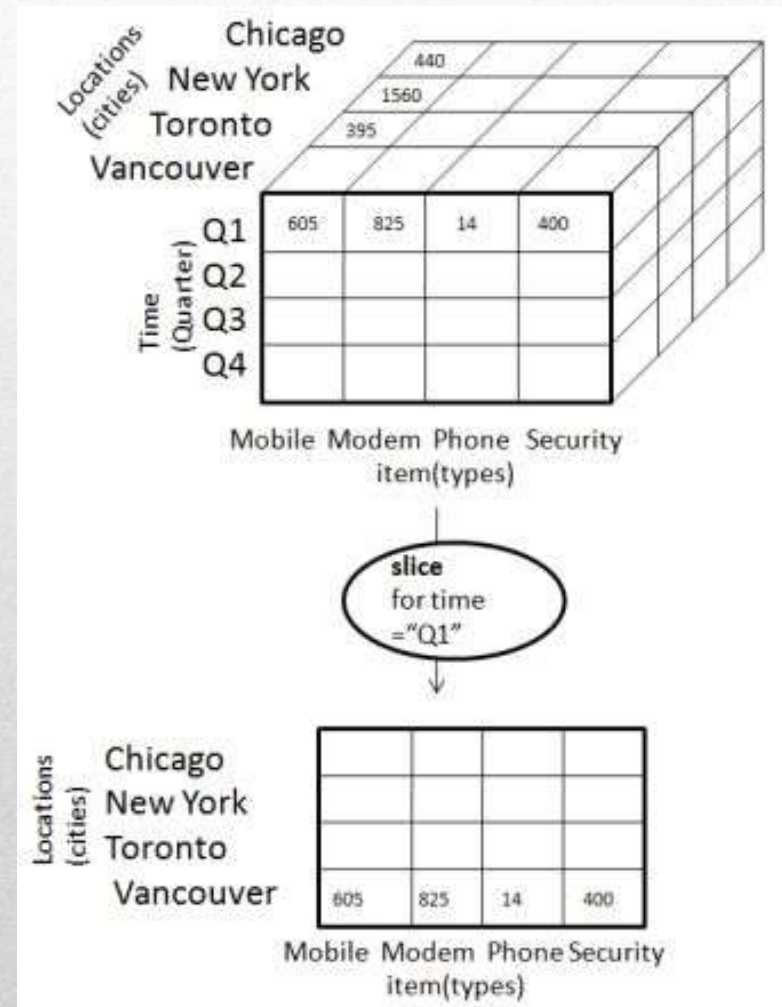
- Drill-down
 - ✓ Is the reverse operation of roll-up. It is performed by either of the following ways –
 - ✓ By stepping down a concept hierarchy for a dimension
 - ✓ By introducing a new dimension.



OLAP operations

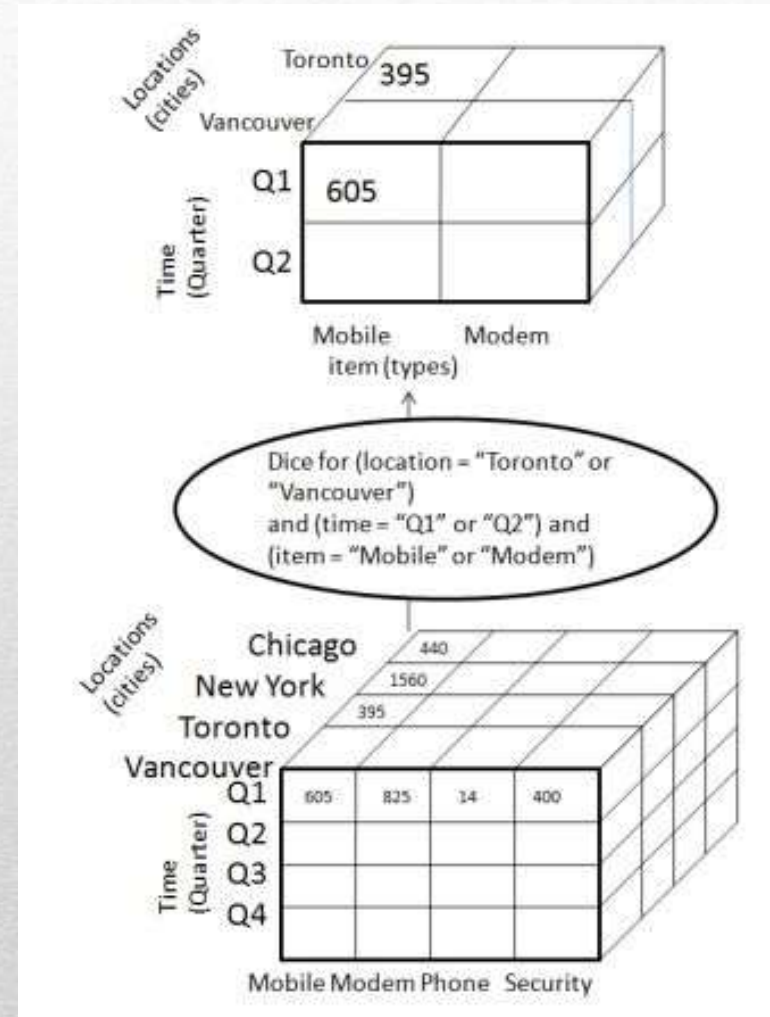
■ Slice

- ✓ The slice operation selects one particular dimension from a given cube and provides a new sub-cube.



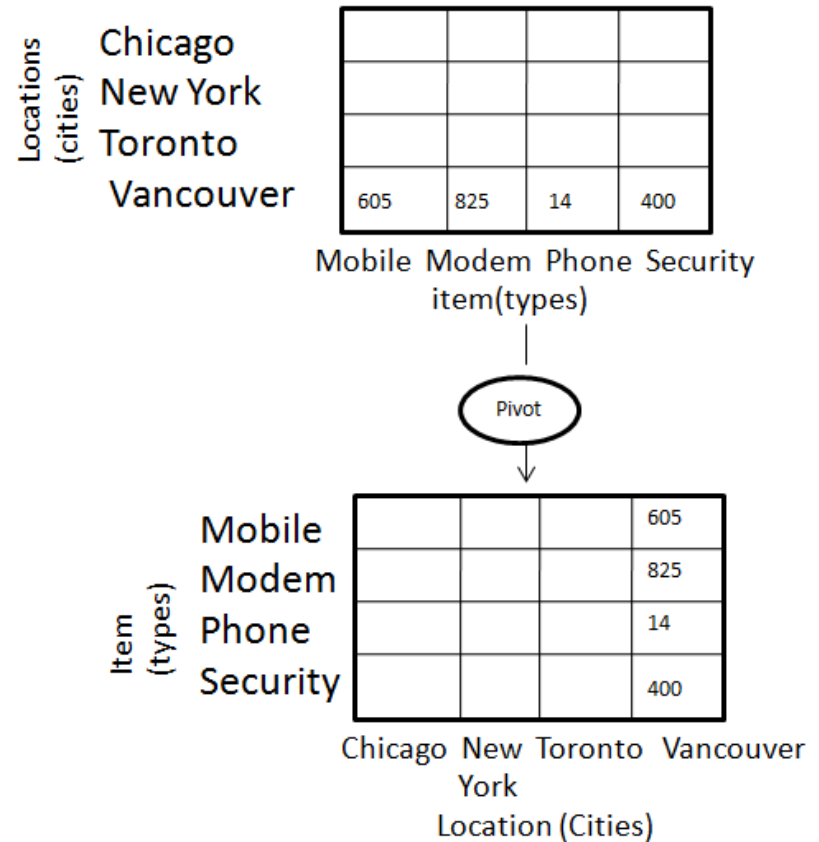
OLAP operations

- Dice
 - ✓ Dice selects two or more dimensions from a given cube and provides a new sub-cube.



OLAP operations

- **Pivot (rotate)**
 - ✓ It rotates the data axes in view in order to provide an alternative presentation of data.



Partitioning Strategy

- ❖ Partitioning is done to enhance performance and facilitate easy management of data by partitioning each fact table into multiple separate partitions.
 - **For easy management,**
 - ✓ This huge size of fact table is very hard to manage as a single entity. Therefore it needs partitioning.
 - **To assist backup/recovery,**
 - ✓ Partitioning allows us to load only as much data as is required on a regular basis.
 - **To enhance performance**
 - ✓ By partitioning the fact table into sets of data, the query procedures can be enhanced.

Partitioning Strategy

❖ Partitioning by Time

- The fact table is partitioned on the basis of time period.
- Each time period represents a significant retention period within the business.
- **For example**, if the user queries for **month to date data** then it is appropriate to partition the data into monthly segments.

❖ Partition on a Different Dimension

- The fact table can also be partitioned on the basis of dimensions other than time such as product group, region, supplier, or any other dimension.
- **For example**, partition the fact table into regional partitions, this cause the queries to speed up because it does not require to scan information that is not relevant.

Partitioning Strategy

❖ Partition by Size of Table

- When there are no clear basis for partitioning the fact table on any dimension, then we should partition the fact table on the basis of their **size**.
- This partitioning is complex to manage, it requires metadata to identify what data is stored in each partition.

❖ Partitioning Dimensions

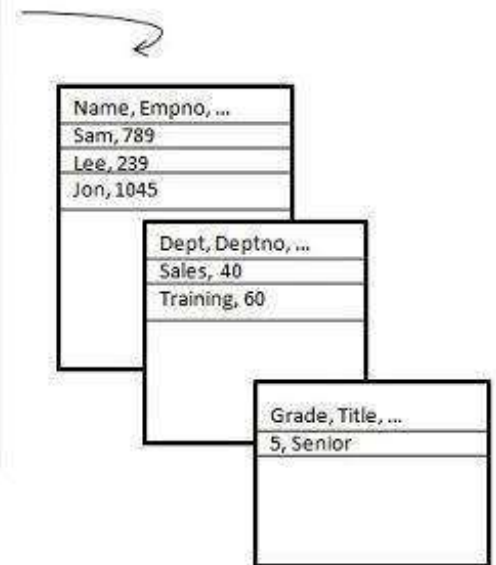
- If a dimension contains large number of entries, then it is required to partition the dimensions.
- If we need to store all the variations in order to apply comparisons, that dimension may be very large.

Partitioning Strategy

❖ Vertical Partition

- Splits the data vertically.
- Vertical partitioning can be performed in the following two ways :
 - Normalization
 - Row Splitting

Name,Empno,...	Dept,deptno,...	Grade, Title,...
Sam, 789	sales, 30	5,Senior, ...
Lee, 239	Training, 60	5, Senior, ...
Jon, 1045	Sales, 40	5, Senior, ...



Partitioning Strategy

❖ Vertical Partition – Example

■ Before normalization

Product_id	Qty	Value	sales_date	Store_id	Store_name	Location	Region
30	5	3.67	3-Aug-13	16	sunny	Bangalore	S
35	4	5.33	3-Sep-13	16	sunny	Bangalore	S
40	5	2.50	3-Sep-13	64	san	Mumbai	W
45	7	5.66	3-Sep-13	16	sunny	Bangalore	S

■ After normalization

Store_id	Store_name	Location	Region
16	sunny	Bangalore	W
64	san	Mumbai	S

Product_id	Quantity	Value	sales_date	Store_id
30	5	3.67	3-Aug-13	16
35	4	5.33	3-Sep-13	16
40	5	2.50	3-Sep-13	64
45	7	5.66	3-Sep-13	16

Process Managers

- ❖ There are three different types of process managers : Load manager, Warehouse manager, and Query manager

1. Load Manager:

- Extract data from the source system.
- Fast load the extracted data into temporary data store.
- Perform simple transformations into structure similar to the one in the data warehouse.

Process Managers

2. Warehouse Manager

- Analyzes the data to perform consistency and referential integrity checks.
- Creates indexes, business views, partition views against the base data.
- Generates new aggregations and updates the existing aggregations.
- Generates normalizations.
- Transforms and merges the source data of the temporary store into the published data warehouse.
- Backs up the data in the data warehouse.

3. Query Manager

- The query manager is responsible for directing the queries to suitable tables.
- It presents the data to the user in a form they understand.
- It schedules the execution of the queries posted by the end-user.

❖ Data warehousing Tools:

- Amazon Redshift
- Google BigQuery
- Snowflake
- Xplenty
- Microsoft Azure
- Oracle 12c
- PostgreSQL
- Teradata
- SAP HANA
- DB2 Warehouse
- Exadata
- Netezza
- MariaDB